



Rust: Create a user and add it to a group on Windows

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rust

note

▼ 其他 2 个属性



The following snippet may be incomplete since I copied and pasted it from a large project. I should have created a snippet!

I'd recommend this excellent resource [🔗 netuser-rs](#). It is a rust-based cli/library to manage Windows users and groups.

My solution is built on top of it but is slightly different: I used `USER_INFO_2` instead of `USER_INFO_1`, which allows for the full name of the user, and `LOCALGROUP_MEMBER_INFO_3`, instead of `LOCALGROUP_MEMBER_INFO_1` which is a simpler API.

```

#[cfg(windows)] fn create_account_win32<S: AsRef<str> + std::fmt::Display>( user_name: S, full_name: S, password: S, group: &str ) -> anyhow::Result<()> { use windows::Win32::NetworkManagement::NetManagement::{ NetUserAdd, UF_NORMAL_ACCOUNT, UF_SCRIPT, USER_ACCOUNT_FLAGS, USER_INFO_2, USER_PRIV_USER, }; // use level 2 i.e. USER_INFO_2 in NetUserAdd // https://learn.microsoft.com/en-us/windows/win32/api/lmaccess/ns-lmaccess-user_info_2 let mut user = USER_INFO_2::default(); let (user_name_w, _v1) = str_to_pwstr(user_name.as_ref()); user.usri2_name = user_name_w; let (password_w, _v2) = str_to_pwstr(password.as_ref()); user.usri2_password = password_w; let (full_name_w, _v3) = str_to_pwstr(full_name.as_ref()); user.usri2_full_name = full_name_w; user.usri2_priv = USER_PRIV_USER; user.usri2_flags = UF_SCRIPT | USER_ACCOUNT_FLAGS(UF_NORMAL_ACCOUNT); user.usri2_acct_expires = u32::MAX; tracing::debug!("Creating user {user:?}"); let status: u32 = unsafe { NetUserAdd(None, 2, &user as *const _ as _, None) }; if status == 0 { println!("✓ Successfully created account."); } else { eprintln!("✗ Failed to create account!"); } // Now add user to a local group. This group must already exist in the system. manage_group_user_s_win32(user_name.as_ref(), group, false)?; Ok(()) } /// Add user to a specified local group. /// If the `deletion` param is `true` - delete the specified user account from the local /// group. #[cfg(windows)] unsafe fn manage_group_users_win32( username: &str, groupname: &str, deletion: bool ) -> anyhow::Result<()> { use windows::Win32::NetworkManagement::NetManagement::{ NetLocalGroupAddMembers, NetLocalGroupDelMembers, LOCALGROUP_MEMBERS_INFO_3, }; let (wide_groupname_nul, _buf) = str_to_pwstr(groupname); let (wide_username_nul, _buf) = str_to_pwstr(username); tracing::debug!("Managing groups of user {username}: group={groupname} , delete={deletion}."); let group_members = LOCALGROUP_MEMBERS_INFO_3 { lgrmi3_domainandname: wide_username_nul, }; let rc = if deletion { NetLocalGroupDelMembers( None, wide_groupname_nul, 3, &group_members as *const _ as *const u8, 1 ) } else { NetLocalGroupAddMembers( None, wide_groupname_nul, 3, &group_members as *const _ as *const u8, 1 ) }; if rc != 0 { anyhow::bail!("Failed to manage user group. Error {rc}."); } Ok(()) } /// Convert &str into PWSTR (null terminated). pub fn str_to_pwstr(s: &str) -> (PWSTR, Vec<u16>) { let mut encoded = s.encode_utf16().chain([0u16]).collect::<Vec<u16>>(); (PWSTR(encoded.as_mut_ptr()), encoded) }

```